

Challenge: Bumper Bowling

Points: 200 • Estimated Time: 45 Minutes

Challenge Scope

The robot must accelerate down a lane and knock over a formation of lightweight plastic bowling pins.

Instructions & Engineering Hints

- **Objective:** Design and program your robot to accelerate down a designated lane, build up maximum momentum, and knock over a formation of lightweight plastic bowling pins at the far end.
- **Structural Considerations:** To achieve high impact force, consider your robot's weight distribution and wheelbase stability. A wider base helps prevent veering off course, while a well-placed front bumper can distribute the impact evenly across the pin formation.
- **Programming Strategy:** Experiment with gradual acceleration profiles rather than an instant full-power burst to maintain traction on the floor surface. Use distance or touch sensors if you need to measure your stopping distance, or time-based routines for consistent high-speed runs.
- **Pro Tip:** Test your approach with different approach angles. Sometimes hitting dead-center yields a different result than striking slightly off-center to cause a chain reaction among the pins.